

REVIVE

4-LFO MIDI CC Generator

MINDBURNER MIDI PRODUCTS

Hand-built in Belfast

User Guide · Firmware v2.3

Overview

REVIVE is a standalone 4-LFO MIDI CC generator. Each of its four LFOs runs independently, with its own shape, speed, depth, MIDI channel, CC number and output routing — free-running at its own rate, or synced to an incoming MIDI clock.

REVIVE does **not** generate its own clock. For clock-synced LFOs, feed it MIDI clock from a Mindburner Masterclock or any other clock source into MIDI IN — free-running LFOs work with no clock at all.

Hardware

Connector	Function
MIDI IN	Clock, transport, and any MIDI to be merged onto the outputs (full thru)
MIDI OUT A	Hardware UART — shares timing with MIDI IN
MIDI OUT B	Hardware UART, independent
MIDI OUT C	Software-generated UART (bit-banged, 31250 baud)
MIDI OUT D	Software-generated UART (bit-banged, 31250 baud)

Every incoming MIDI message — notes, clock, CC, anything — is merged onto all four outputs, alongside whatever CC data the LFOs are sending. REVIVE can sit inline in a live MIDI chain without interrupting it.

What each LFO controls

Every LFO independently sets: its waveform **Shape**, whether it runs free or clock-synced (**Mode**), its rate or clock division (**Speed**), how much smoothing is applied to the value it sends (**Glide**), the CC's swing/depth and centre point (**Swing** / **Mid**), which MIDI **Channel** and **CC** number it sends on, and which physical output (or pair of outputs) it's routed to.

Persistence

REVIVE stores up to 8 complete 4-LFO patches in named **Save Slots**. It boots from Slot 1 — whatever is saved there is what loads at power-up. A small ***** in the header means the current patch has changes not yet saved into any slot.

The factory-default patch (loaded the first time you power up a new REVIVE, before you've saved anything) has all 4 LFOs on, each sending on its own MIDI channel — LFO1 on channel 1, LFO2 on channel 2, and so on through LFO4 on channel 4.

Main Screen Controls

REVIVE's main screen can display in two modes — **Graphical** (icon/dial based, one LFO fills the screen, and the default view) or **Text** (a row-by-row list, described below). Both use the exact same controls and the exact same 9 fields underneath — only the drawing differs. See the Graphical Mode page for details.

In Text mode, the main screen shows one LFO at a time, selected via the four tabs (LFO1–LFO4) across the top. The currently focused field is marked with a slowly pulsing **>** cursor.

Control	Action	Result
Rotate	Turn	Decrease / increase the focused field's value (CW = increase, CCW = decrease)
ENC_BTN	Short press	Move to the next field — mirrors C, handy for single-handed, knob-only operation
D	Short press	Switch to the next LFO (1→4→1)
D	Hold	Context-sensitive: on **Ch** or **CC**, arms MIDI Learn; on **Mode**, toggles retrigger-on-MIDI-Start (flashes “LFOn RETRIG ON/OFF”); anywhere else, turns the current LFO on/off (flashes “LFOn ON/OFF”)
C	Short press	Move to the next field (9 fields, in order: Shape → Mode → Speed → Glide → Swing → Mid → Ch → CC → Sends to)
C	Hold	Go to the Save Slots screen. Nothing is saved automatically — this only navigates

There's no repeat-acceleration logic to configure — a physical knob naturally moves further per second the faster you turn it, so a fast spin covers more ground on its own.

Global Mute

Hold **C + D** together (~1.2s) to instantly mute all 4 LFOs' CC output at once — a quick kill switch, separate from each LFO's own **Off (bypass)** Sends to option below. The header tints red and flashes “ALL OUTPUTS MUTED” (or “UNMUTED”) to confirm. Hold C+D again to release it.

MIDI thru (notes, clock, anything passing through) is **not** affected — only the LFOs' own CC output is blocked. This is never saved — REVIVE always powers on un-muted, so a mid-gig power cycle can't silently leave outputs muted without you knowing.

Field reference

Field	What it does
Shape	The LFO waveform — see the Waveforms table below
Mode	Free-running (its own Hz rate) or Clock-synced (locked to a musical division of incoming MIDI clock)
Speed	Free-running: rate in Hz (0.002–20 — the slow end is a genuinely glacial ~8.3 minute sweep). Clock-synced: division from 1/32 note to 4 bars
Glide	0–9. Smooths the sent CC value — 0 is instant, higher values widen the smoothing time (up to ~480ms)
Swing	0–63. How far the LFO swings the CC value away from Mid
Mid	0–127. The CC value the LFO swings around
Ch	MIDI channel (1–16) this LFO's CC messages are sent on
CC	CC number (0–119) this LFO controls
Sends to	Which output this LFO's CC goes to: A, B, C, D, a pair (A&B, B&C, C&D), All outputs, or **Off (bypass)** — the LFO keeps running (Send bar and all) but nothing actually reaches a wire

Waveforms

Shape	Description
Sine	Smooth rise and fall
Triangle	Straight-line rise to a peak at the midpoint, then straight-line fall
Inv Triangle	Mirror of Triangle — starts high, dips to a trough at the midpoint, rises back (a “V” shape)
Saw Up	Linear ramp from low to high, then an instant drop
Saw Down	Linear ramp from high to low, then an instant jump
Square	Alternates instantly between low and high
Random (S&H)	A new random value each cycle, held steady until the next (sample & hold)

Graphical Mode

Graphical Mode is REVIVE's default main-screen view: one LFO fills the whole screen, shown with icons and dials instead of the text rows described on the previous page. It uses the exact same fields, same values, and mostly the same buttons — it's a different way of looking at the same 9 settings, not a different feature set.

Switching between Graphical and Text mode

Hold **ENC_BTN** (~1.2s)	Switches modes right now, either direction. The header flashes “GRAPHICAL MODE” or “TEXT MODE” in cyan to confirm.
Hold **ENC_BTN** while powering on	Boots into Text mode for that session only — not saved anywhere. A normal boot next time returns to Graphical.

What's on screen

Element	Field / meaning
Header	Shows “LFOn: FIELD” for whichever field is currently focused (e.g. “LFO2: SWING”), instead of the REVIVE logo. The unsaved-changes ***** still blinks here when it applies
Four dots (top)	Which of the 4 LFOs is selected — replaces the tabs
Waveform icon	Shape
Dial + number	Speed — sweeps like a volume knob (empty at ~7:30, full at ~4:30, through 12 o'clock); the exact Hz or clock division always prints below it
Send bar	The live CC value currently being sent, updating continuously — the same readout Text mode shows at the bottom of its screen
Horizontal bar	Swing and Mid together — the shaded band is Swing (how far), the white marker is Mid (centred where)
FREE / SYNC / WAITING	Mode — WAITING (in red) replaces SYNC when this LFO is clock-synced but no MIDI clock has been seen recently, matching Text mode's “NO CLOCK” warning
Coloured chip	Sends to (output routing)
Row of dots	Glide — filled left-to-right up to the current value (0–9)
CH / CC numbers	Channel and CC number, shown plainly — no icon beats reading the actual number

Navigation

Rotate, C and D all do the same jobs as in Text mode — rotating adjusts the focused field, D switches LFO or toggles context actions, C-hold still goes to Save Slots. The one real difference: **C's cycling order follows the on-screen layout** in Graphical Mode (Shape → Speed → Swing → Mid → Mode → Output → Glide → Ch → CC) rather than Text mode's row order — worth knowing if you're used to counting presses out of habit.

A pulsing rectangle outline (in place of the **>** cursor) shows which of these zones is currently focused.

MIDI Learn

MIDI Learn lets you set an LFO's **Ch** and **CC** fields directly from an incoming controller message, instead of dialling them in by hand.

To use it

1	On the main screen, press C repeatedly to focus the **Ch** or **CC** field of the LFO you want to set
2	Hold D. The header shows **“LFOn LEARN: send CC”** in yellow
3	Move a control on your MIDI keyboard/controller that sends a CC message into MIDI IN
4	REVIVE grabs the channel and CC number automatically — the header flashes **“LFOn LEARNED”** in green

MIDI Learn auto-cancels after 8 seconds if nothing arrives.

Note: Learn only recognizes a CC message that includes its own status byte. Some controllers use MIDI's “running status” shorthand for repeated CCs, omitting the status byte on messages after the first. If a control doesn't seem to register, move a different control on the same device first to force a fresh status byte, then move the one you want to learn.

Retrigger on MIDI Start

With the **Mode** field focused, hold D to toggle whether this LFO's phase snaps back to zero whenever a MIDI Start message arrives — shown as an “R” suffix next to the Mode value, and confirmed with a brief “LFOn RETRIG ON/OFF” flash in the header. Useful for keeping an LFO's sweep locked to the start of playback.

Save Slots

The Save Slots screen holds up to 8 complete 4-LFO patches. REVIVE boots from **Slot 1** — saving there is how you set what loads at power-up.

Browse mode

Control	Action	Result
Rotate	Turn	Previous/next slot
ENC_BTN	Short press	Save the current patch into the selected slot (mirrors C; confirm-gated, same as C)
ENC_BTN	Hold	Enter rename mode
C	Short press	Save the current patch into the selected slot (2nd tap within 3s to confirm, unless empty)
C	Hold	Back to the main screen
D	Short press	Load the selected slot (2nd tap within 3s to confirm, unless empty)
D	Hold	Clear the selected slot (2nd hold within 3s to confirm, unless already empty)

Save, Load and Clear all require a confirming second press within 3 seconds — skipped only when the slot is already empty, since there's nothing at stake.

Rename mode (hold ENC_BTN to enter or exit)

Control	Action
Rotate	Step the character at the cursor forward/backward (space, A–Z, 0–9)
D	Confirm this letter, move the cursor to the next character (wraps after the last)

No holding needed to step through letters or move on — only the ENC_BTN hold to enter or leave rename mode itself.

Updating the Firmware

REVIVE's firmware is updated over USB using a browser-based installer — no separate software to install, and no need to open the case.

mindburners.github.io/REVIVE-QUAD-LFO

The same installer is also linked from mindburner.co.uk/page/fw-updatesguides, alongside firmware and guides for every other current Mindburner device.

1	Connect REVIVE to your computer via USB
2	Open the installer link above in **Chrome** or **Edge** — the tool relies on WebSerial, which Safari and Firefox don't support
3	Click the install button and select REVIVE's serial port when your browser prompts for it
4	Click **Install** and wait for it to finish — don't unplug the USB lead or close the browser tab while it's flashing
5	REVIVE reboots automatically into the new firmware once the install completes

The page shows the latest available firmware version and a short changelog before you install.

Note: REVIVE uses a Silicon Labs CP2102 USB-to-serial chip. Most systems recognize it automatically; on older Windows installs, if the port doesn't appear, installing the CP210x driver from Silicon Labs' website resolves it.

If you want to be safe, note down or photograph any patches you'd hate to lose before updating — REVIVE has no way to export a patch to your computer, so there's currently no other way to back one up.

Tips & Troubleshooting

Getting four LFOs to feel less locked-together

Turning an LFO on via D-hold gives it a fresh quadrature phase offset based on its tab position (0°/90°/180°/270°), so four LFOs starting together spread out across their cycle instead of moving in lock-step. Using four different Shapes reinforces this further.

Smoothing out stepped or jumpy CC data

Raise **Glide** on an LFO if the receiving synth or effect reacts too abruptly to Random (S&H) or Square shapes — higher settings widen the smoothing time up to roughly half a second.

“Waiting for MIDI clock...”

Shown when an LFO's Mode is set to Clock-synced but no MIDI clock has been seen recently. Feed MIDI clock into MIDI IN from a Masterclock or DAW — or switch that LFO's Mode to Free-running if no clock source is available.

Where patches actually live

There is no separate “live” patch storage — REVIVE boots from whatever is saved in Slot 1. Editing on the main screen only changes what's currently loaded in memory; it's not persisted until you explicitly save it into a slot on the Save Slots screen.